RAHEE

Weekly Project Progress Report

Project	RAHEE — an online digital version of a travel board game
Team	Team 31 — 5 members
Current stage	We spent our first week studying the physical game's rulebook and mechanics in detail, including two in-person playtesting sessions with the project's Point of Contact to resolve rule interpretations left ambiguous by the written manual. A single-device (hotseat) prototype was then implemented and tested against the official rulebook, covering the complete data-driven rules engine, board rendering and gameplay loop. The architecture for the real-time, multi-device online version has since been finalised, and its implementation is now under way.

1. Tasks — Completed, In Progress and Planned

Completed

- Spent one week studying the official rulebook and the printed board, including two in-person playtesting sessions with the project's Point of Contact, to confirm rule interpretations that were ambiguous in the written manual.
- Initial project setup completed: a dependency-free, browser-based application built using plain ES modules with no build step, with the team's repository and development workflow established.
- The complete board encoded as a single structured data file, comprising 101 cities, 209 routes (road, flight and ferry), 51 tourist places, 6 playable characters with 36 destination goals in total, and the full pricing and scoring table specified in the rulebook.
- A rules engine implemented independently of any board-specific knowledge, covering hand dealing per player count, card-based movement with its associated travel costs and surcharges, the four memory suits, tourist-place visits, the complete road-trip mechanic (invitation, shared cost, shared-tourism discount and drop-off), all three permitted uses of an Ace, detection of the end-of-game trigger, and full score computation.
- The game board rendered directly from the data file over the supplied map artwork, including pan/zoom navigation, drawn routes, player tokens, and highlighting of legal moves.
- A complete single-device (hotseat) gameplay loop implemented: player setup, a private handover screen between turns, the turn-action panel, and a final scoreboard with a transparent score breakdown.
- Save and resume functionality implemented for an in-progress single-device game.
- An automated test suite built, covering random full-game simulation (checking deck composition, rule invariants, and the rulebook's own worked examples) and a headless-browser walkthrough that drives a complete game through real pointer input.

- The architecture for the real-time, multi-device online version finalised, including the client–server split, the WebSocket-based real-time connection, server-side redaction of private player state, room-based sessions, and the design of the public/private chat feature.

In Progress

- The server process: the room manager, the authoritative rules engine running server-side, and the per-player state-redaction layer that determines what each connected device is permitted to see.
- The real-time connection layer between each player's device and the server, carrying turn actions, movement and chat messages over a single connection per device.
- Destination-based route planning: given a played card, the system computes every route to a chosen destination within the card's remaining movement allowance, keeping this candidate-route computation private to the active player until a route is confirmed.
- Public and private chat: a room-wide public channel that every connected player can see, for general discussion and coordinating trips, alongside a private one-to-one channel that lets two players negotiate individually — for example, negotiating a road-trip invitation — without the conversation being visible to anyone else in the game. Both channels will run over the same real-time connection as the game state itself, with server-side message sanitisation and rate-limiting to prevent spam or malicious input, consistent with the private-by-default handling already designed for a player's hand and route plans.
- Room creation and joining by a short, shareable room code, so that a game can be set up without any account system.
- Adapting the existing board rendering and turn-action interface so that it reacts to state pushed from the server, rather than calling the rules engine locally.

Planned

RAHEE is being built as a cross-platform online multiplayer game: every player will be able to play from their own device, phone or laptop, within the same game session, rather than sharing one screen, with public and private chat available throughout for discussion during play.

- Support for both phone and laptop screens, so a player can join and play a full game from whichever device they have.
- Every player playing from their own device within the same game session, in addition to the existing single-device (hotseat) mode for when players share one screen.
- Public and private chat available throughout a game for discussion, in addition to their planned use in road-trip negotiation.
- Reconnection handling (token-based rejoin after a dropped connection) and server-side save/resume.
- Responsive, touch-first layout for phones.
- Continuous integration: linting, automated tests, and a security scan on every change.
- Deployment to a stateful hosting environment.
- AI players to fill empty seats (a later-phase requirement).
- Complete end-to-end multiplayer testing and QA defect tracking.

2. Problems / Issues

Reported by the PoC

- Movement is currently performed one step at a time, by repeatedly clicking an adjacent city, rather than by choosing a destination directly. The intended behaviour is that a player should be able to click any city on the board and immediately see whether it is reachable within the current card's movement allowance, showing the number of steps required if it is reachable (for example, up to 6 steps for a card worth 6), or clearly indicating that the destination is not possible if it exceeds the card's value. This is captured in the finalised architecture (see the destination-based route planning item under “In Progress” above) and is being implemented as part of the online version.

Additional issues

- The connection data for the printed board (which cities join, route types, exact step counts and surcharges) has not yet been supplied by the client (CCL). Every route currently uses a placeholder value of one step, traced from the printed artwork, pending the official data; this will need to be swapped in once received.
- The current save/resume feature stores game state in the browser on a single device. This approach does not extend to the multi-device online version and will need to be replaced by server-side persistence as part of the ongoing server implementation.
- Undo is currently limited to reversing the last single movement step, rather than a whole turn action; extending this remains an outstanding item.
- A small number of city markers on the board (for example, where a marker was obscured by illustration or a text label) were not found by the automatic marker-detection step and had to be placed manually. All positions have since been verified by eye, but this is a limitation to keep in mind if the board artwork changes.

3. Time Spent by Each Team Member — Previous Week

Hours are divided equally across the team for the previous week's work, covering the task areas below.

Name	Task(s)	Approx.. Hours/week
Tanuj Sharma	Rules engine implementation (movement, costs, memories, road trips, scoring)	3
Sai Gawali	Board data file and data-driven board rendering (map, routes, tokens)	3
Siddhesh Patil	Application screens (setup, private handover, sidebar, scoreboard)	3
Mukesh Dewangan	Automated testing (random-game simulation and headless-	3

	browser walkthrough)	
Kshitij Kasodkar	Online multiplayer architecture design and documentation	3

4. Last Meeting with the PoC

Date of last meeting	9 September 2026 (Wednesday)
Meeting cadence	Weekly, every Wednesday. The previous week's meeting (16 September 2026) was cancelled due to the PoC's unavailability.
Mode	Offline

Key discussion / action items:

- The game must run across multiple systems, with one device per user — a correction to the original approach, where a single device was shared by multiple users.
- While a player is choosing their route, the candidate paths under consideration are not shown to other players; only once the player confirms a route does the resulting position become visible to everyone.
- The active player may invite other players on a road trip; those invited players can see the candidate route only once the active player has agreed to include them.
- A player must be able to select a target city directly: if it is reachable within the current card's maximum number of steps, the system should display the route(s) to it; if it is not reachable within that number of steps (or fewer, as the card allows), the system should indicate that it is not possible.
- Once the system has proposed a route to the current player, the player must be able to adjust it — by dragging a node (city) on the path, or by selecting or deselecting an individual node — rather than only being able to accept or discard the path as a whole.

5. Date of the Current Build

Current build date	17 September 2026
Build / version identifier	Commit 6badcd0, main branch

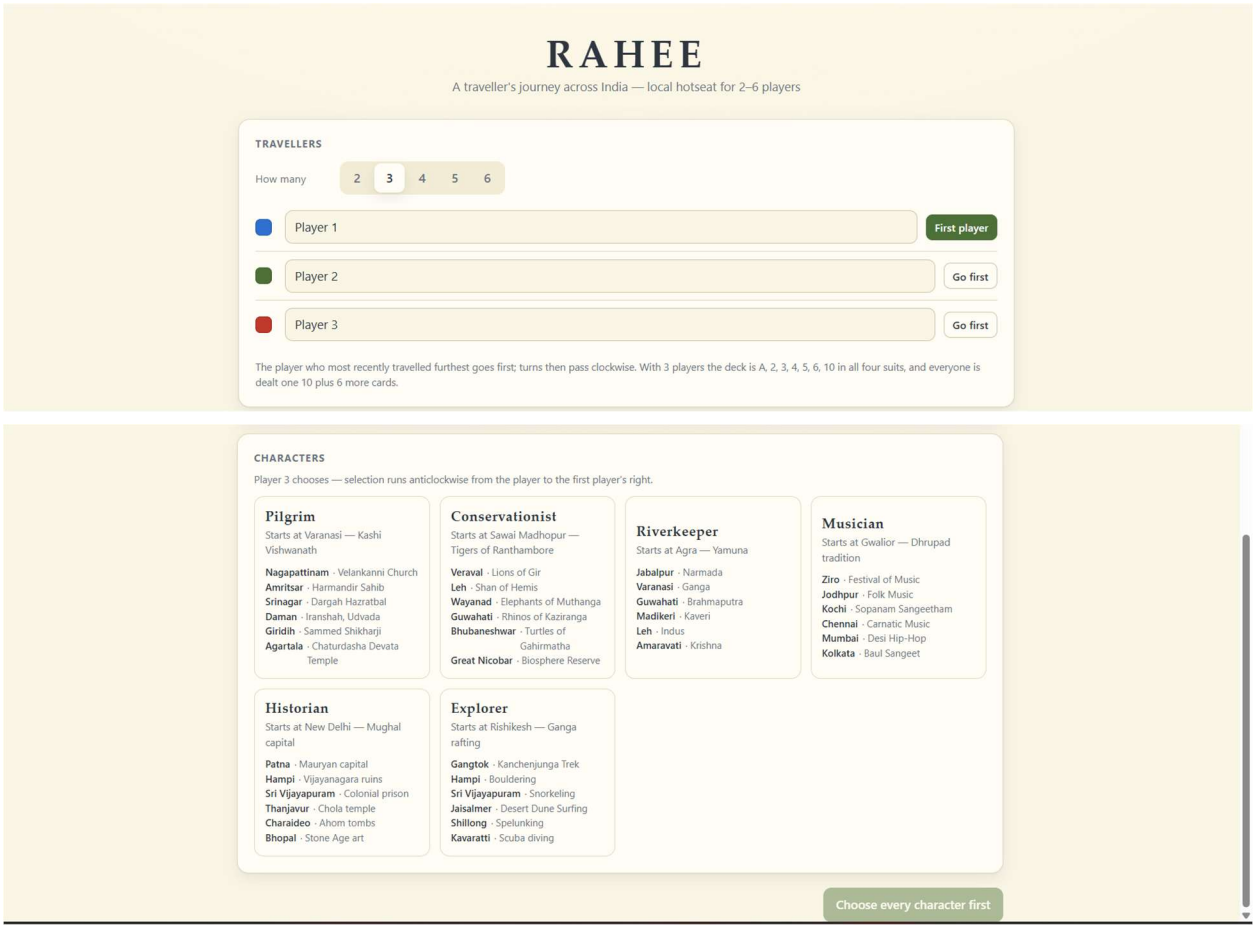
6. Test Results on the Current Build

Tests are run against the rules engine: an automated random-game simulation, plus the rulebook's own worked examples, together with a headless-browser walkthrough of a full single-device game.

Total tests	82 (80 randomly generated full-game simulations, each checking multiple rule invariants, plus 2 fixed rulebook worked-example checks)
Unsuccessful tests	0

Successful tests	82

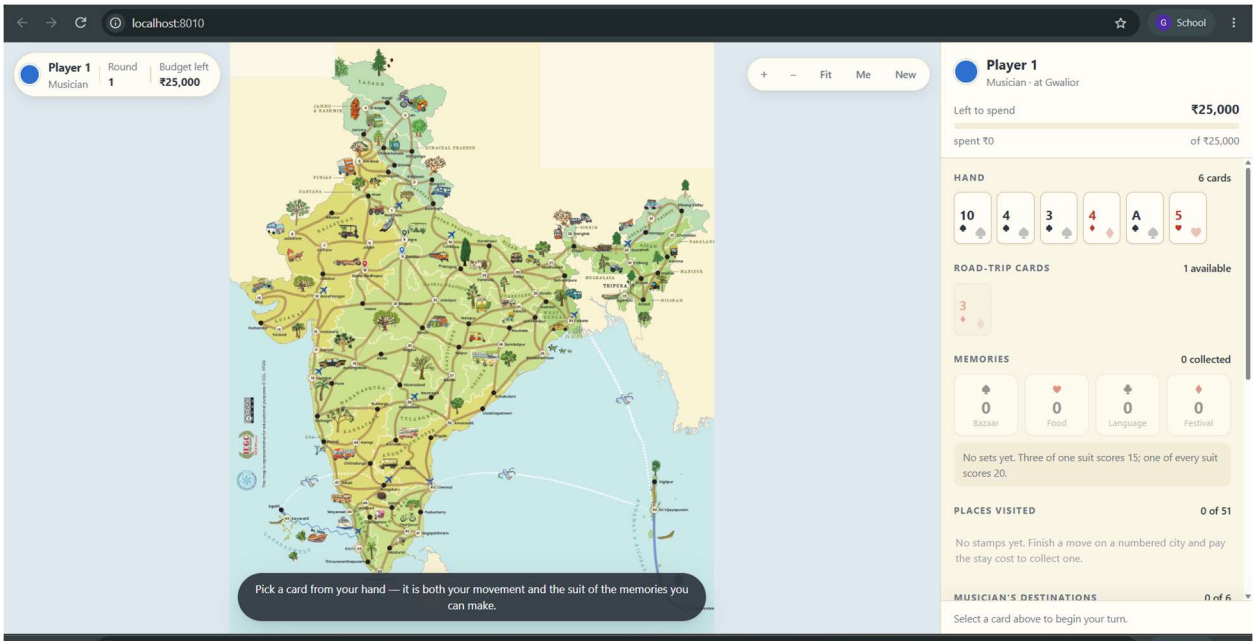
7. Evidence

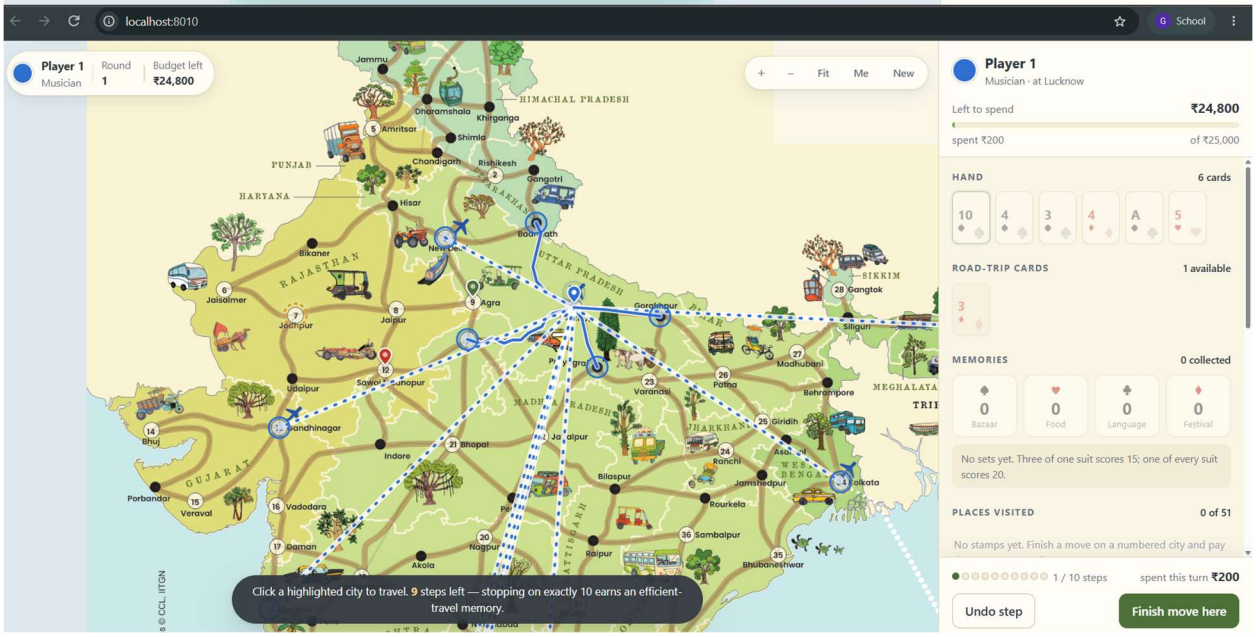


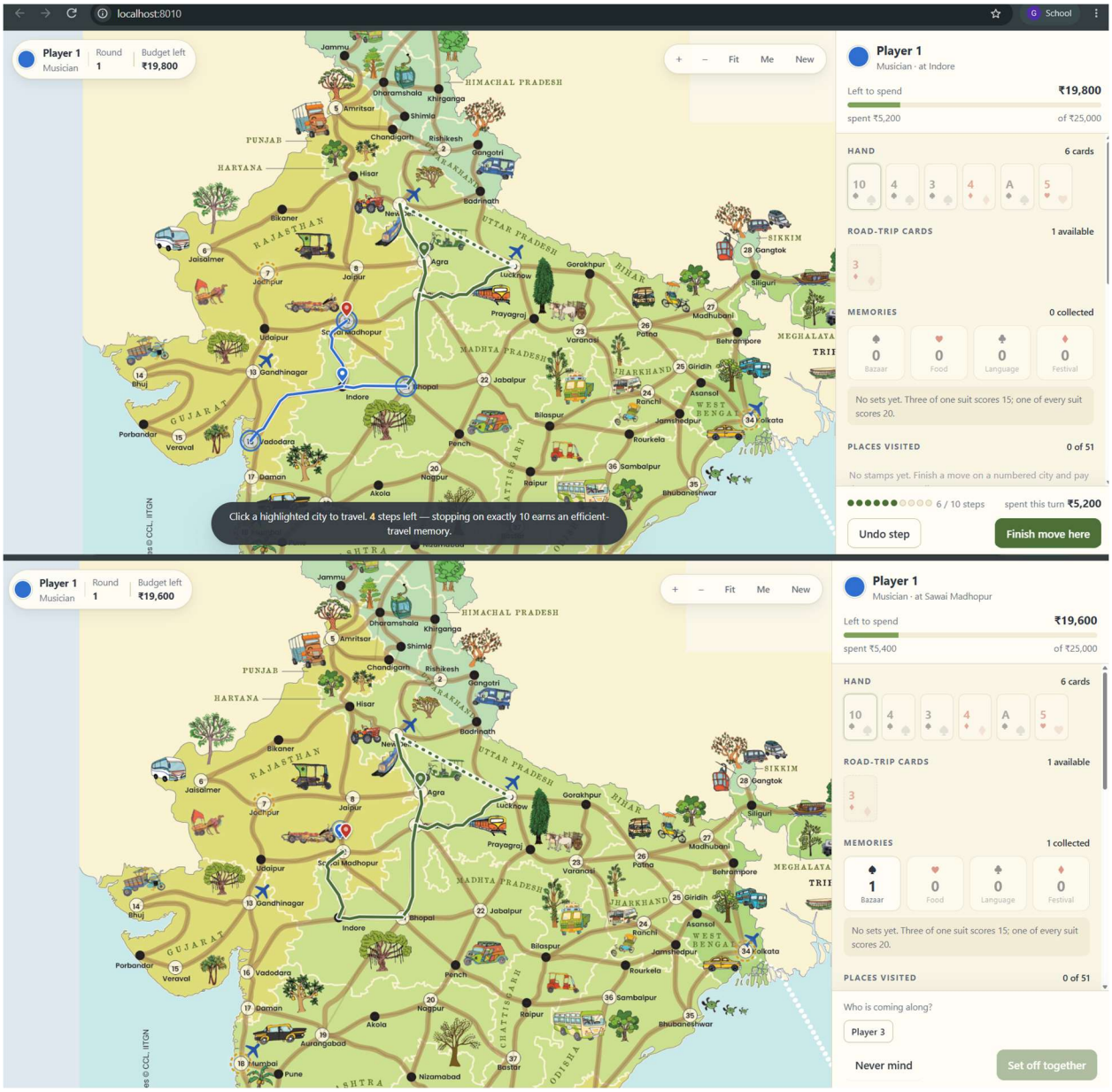
Player 2, lay a road-trip card

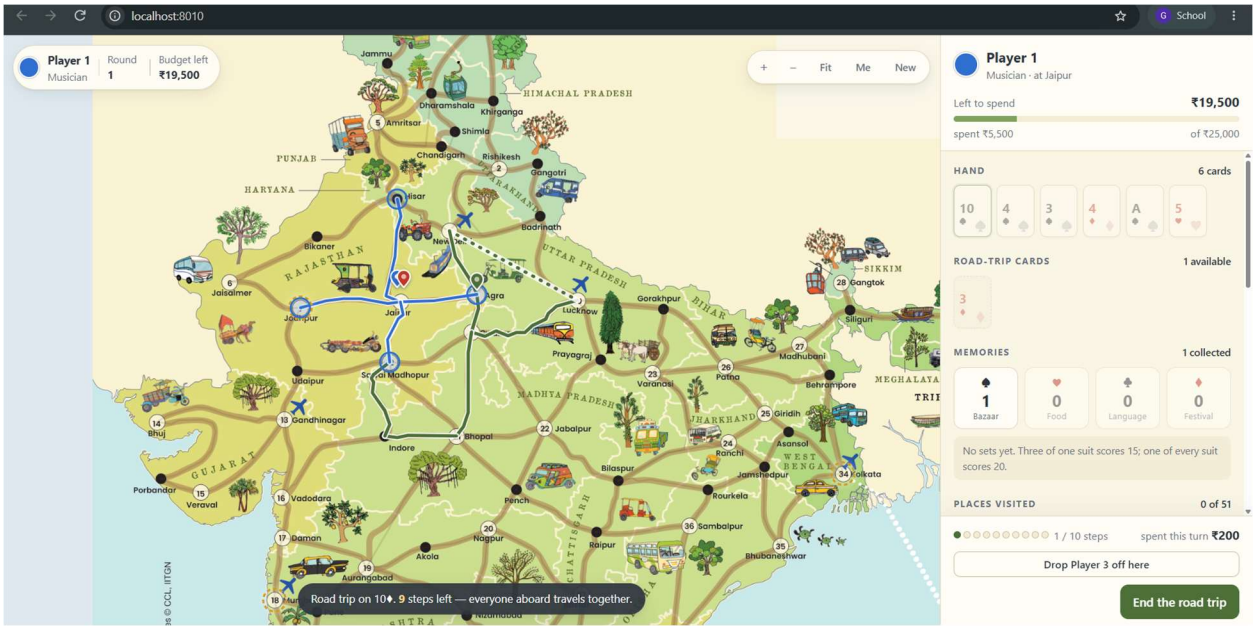
One card from your hand goes face-up in front of you. Others can flip it to take you on a road trip — and its value is how far that trip can go. The rest of your hand stays hidden.




localhost:8010
School
Player 1
Musician
Round
1
Budget left
₹25,000
+
−
Fit
Me
New
Player 1
Musician · at Gwalior
Left to spend
₹25,000
spent ₹0
of ₹25,000
HAND
6 cards
10
4
3
4
A
5
ROAD-TRIP CARDS
1 available
3
MEMORIES
0 collected
0
Bazaar
0
Food
0
Language
0
Festival
No sets yet. Three of one suit scores 15; one of every suit scores 20.
PLACES VISITED
0 of 51
No stamps yet. Finish a move on a numbered city and pay the stay cost to collect one.
MUSICIAN'S DESTINATIONS
0 of 6
Pick a card from your hand — it is both your movement and the suit of the memories you can make.
Select a card above to begin your turn.


localhost:8010
School
Player 1
Musician
Round
1
Budget left
₹24,800
+
−
Fit
Me
New
Player 1
Musician · at Lucknow
Left to spend
₹24,800
spent ₹200
of ₹25,000
HAND
6 cards
10
4
3
4
A
5
ROAD-TRIP CARDS
1 available
3
MEMORIES
0 collected
0
Bazaar
0
Food
0
Language
0
Festival
No sets yet. Three of one suit scores 15; one of every suit scores 20.
PLACES VISITED
0 of 51
No stamps yet. Finish a move on a numbered city and pay
Jammu
HIMACHAL PRADESH
Dharamshala
Khirganga
Amritsar
Shimla
PUNJAB
Chandigarh
Rishikesh
Gangotri
HARYANA
Hisar
Badrinath
Bikaner
Jaisalmer
Jodhpur
Jaipur
Agra
Gorakhpur
SIKKIM
Gangtok
Siliguri
Madhubani
Patna
Varanasi
Behrampore
Udaipur
Sawai Madhopur
Gandhinagar
Bhuj
Indore
Bhopal
Giridih
Ranchi
Jamshedpur
Kolkata
Bilaspur
Rourkela
Porbandar
Veraval
Vadodara
Daman
Nagpur
Raipur
Sambalpur
Bhubaneshwar
Akola
Click a highlighted city to travel. 9 steps left — stopping on exactly 10 earns an efficient-travel memory.
1 / 10 steps
spent this turn ₹200
Undo step
Finish move here


localhost:8010
School
Player 1
Musician
Round
1
Budget left
₹19,800
Fit
Me
New
HIMACHAL PRADESH
PUNJAB
HARYANA
RAJASTHAN
UTTAR PRADESH
BIHAR
SIKKIM
MADHYA PRADESH
JHARKHAND
WEST BENGAL
GUJARAT
CHHATTISGARH
ODISHA
Jammu
Dharamshala
Khirganga
Amritsar
Shimla
Chandigarh
Rishikesh
Gangotri
Hisar
Badrinath
Bikaner
Jaisalmer
Jodhpur
Jaipur
Agra
Lucknow
Gorakhpur
Gangtok
Siliguri
Udaipur
Sawai Madhopur
Prayagraj
Varanasi
Patna
Madhubani
Behrampore
Bhuj
Gandhinagar
Indore
Bhopal
Jabalpur
Giridih
Asansol
Ranchi
Jamshedpur
Kolkata
Porbandar
Veraval
Vadodara
Pench
Bilaspur
Rourkela
Daman
Nagpur
Raipur
Sambalpur
Akola
Bhubaneshwar
Aurangabad
Mumbai
Pune
Nizamabad
Bastar
Click a highlighted city to travel. 4 steps left — stopping on exactly 10 earns an efficient-travel memory.
Player 1
Musician · at Indore
Left to spend
₹19,800
spent ₹5,200
of ₹25,000
HAND
6 cards
10
4
3
4
A
5
ROAD-TRIP CARDS
1 available
3
MEMORIES
0 collected
0
Bazaar
0
Food
0
Language
0
Festival
No sets yet. Three of one suit scores 15; one of every suit scores 20.
PLACES VISITED
0 of 51
No stamps yet. Finish a move on a numbered city and pay
6 / 10 steps
spent this turn ₹5,200
Undo step
Finish move here
Player 1
Musician
Round
1
Budget left
₹19,600
Fit
Me
New
Player 1
Musician · at Sawai Madhopur
Left to spend
₹19,600
spent ₹5,400
of ₹25,000
HAND
6 cards
10
4
3
4
A
5
ROAD-TRIP CARDS
1 available
3
MEMORIES
1 collected
1
Bazaar
0
Food
0
Language
0
Festival
No sets yet. Three of one suit scores 15; one of every suit scores 20.
PLACES VISITED
0 of 51
Who is coming along?
Player 3
Never mind
Set off together


localhost:8010
School
Player 1
Musician
Round
1
Budget left
₹19,500
+
−
Fit
Me
New
HIMACHAL PRADESH
PUNJAB
HARYANA
RAJASTHAN
UTTAR PRADESH
BIHAR
SIKKIM
MADHYA PRADESH
JHARKHAND
WEST BENGAL
GUJARAT
MEGHALAYA
Jammu
Dharamshala
Khirganga
Amritsar
Shimla
Chandigarh
Rishikesh
Gangotri
Hisar
Badrinath
Bikaner
Jaisalmer
Jaipur
Agra
Lucknow
Gorakhpur
Gangtok
Siliguri
Prayagraj
Varanasi
Patna
Madhubani
Behrampore
Udaipur
Gandhinagar
Bhuj
Indore
Bhopal
Jabalpur
Giridih
Asansol
Ranchi
Jamshedpur
Kolkata
Bilaspur
Rourkela
Porbandar
Veraval
Vadodara
Daman
Pench
Nagpur
Akola
Raipur
Sambalpur
Bhubaneshwar
Aurangabad
Road trip on 10♦. 9 steps left — everyone aboard travels together.
Player 1
Musician · at Jaipur
Left to spend
₹19,500
spent ₹5,500
of ₹25,000
HAND
6 cards
10
4
3
4
A
5
ROAD-TRIP CARDS
1 available
3
MEMORIES
1 collected
1
Bazaar
0
Food
0
Language
0
Festival
No sets yet. Three of one suit scores 15; one of every suit scores 20.
PLACES VISITED
0 of 51
1 / 10 steps
spent this turn ₹200
Drop Player 3 off here
End the road trip